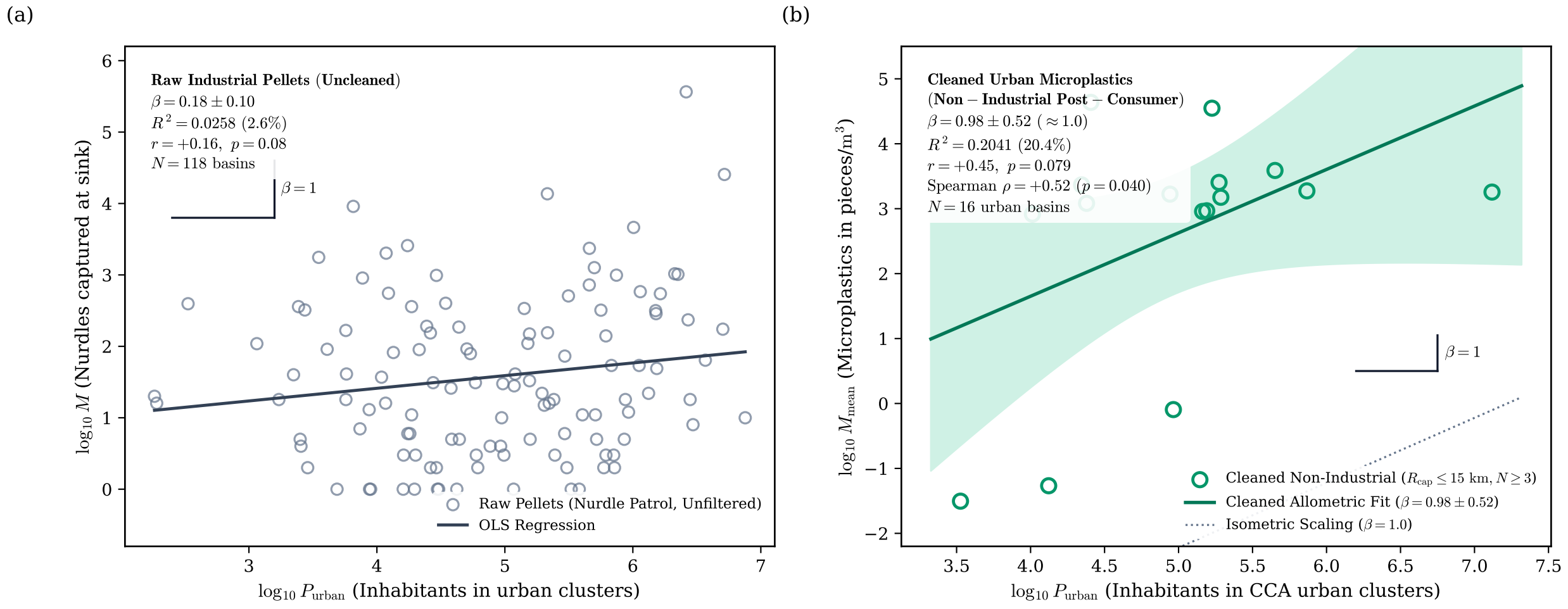

(a)
Raw Industrial Pellets (Uncleaned)
$\beta = 0.18 \pm 0.10$
$R^2 = 0.0258$ (2.6%)
$r = +0.16,\ p = 0.08$
$N = 118$ basins
$\beta = 1$
Raw Pellets (Nurdle Patrol, Unfiltered)
OLS Regression
$\log_{10} M$ (Nurdles captured at sink)
$\log_{10} P_{\text{urban}}$ (Inhabitants in urban clusters)
(b)
Cleaned Urban Microplastics
(Non − Industrial Post − Consumer)
$\beta = 0.98 \pm 0.52$ (≈ 1.0)
$R^2 = 0.2041$ (20.4%)
$r = +0.45,\ p = 0.079$
Spearman $\rho = +0.52$ ($p = 0.040$)
$N = 16$ urban basins
$\beta = 1$
Cleaned Non-Industrial ($R_{\text{cap}} \leq 15$ km, $N \geq 3$)
Cleaned Allometric Fit ($\beta = 0.98 \pm 0.52$)
Isometric Scaling ($\beta = 1.0$)
$\log_{10} M_{\text{mean}}$ (Microplastics in pieces/m^3)
$\log_{10} P_{\text{urban}}$ (Inhabitants in CCA urban clusters)